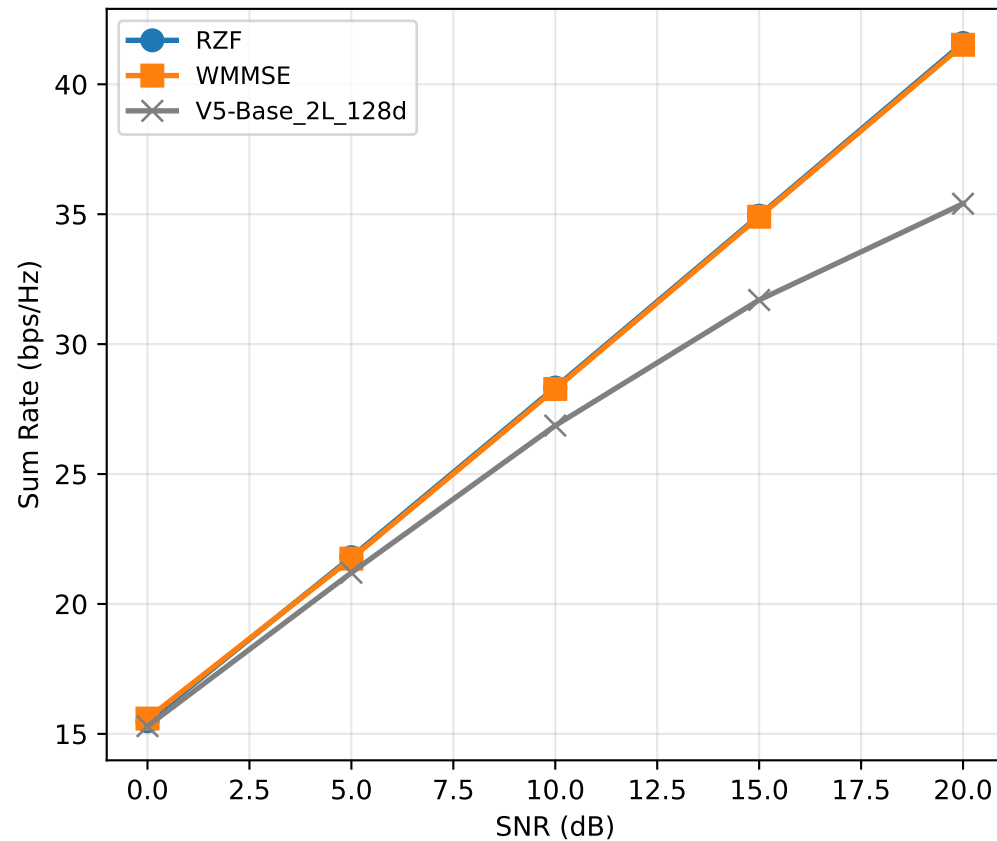
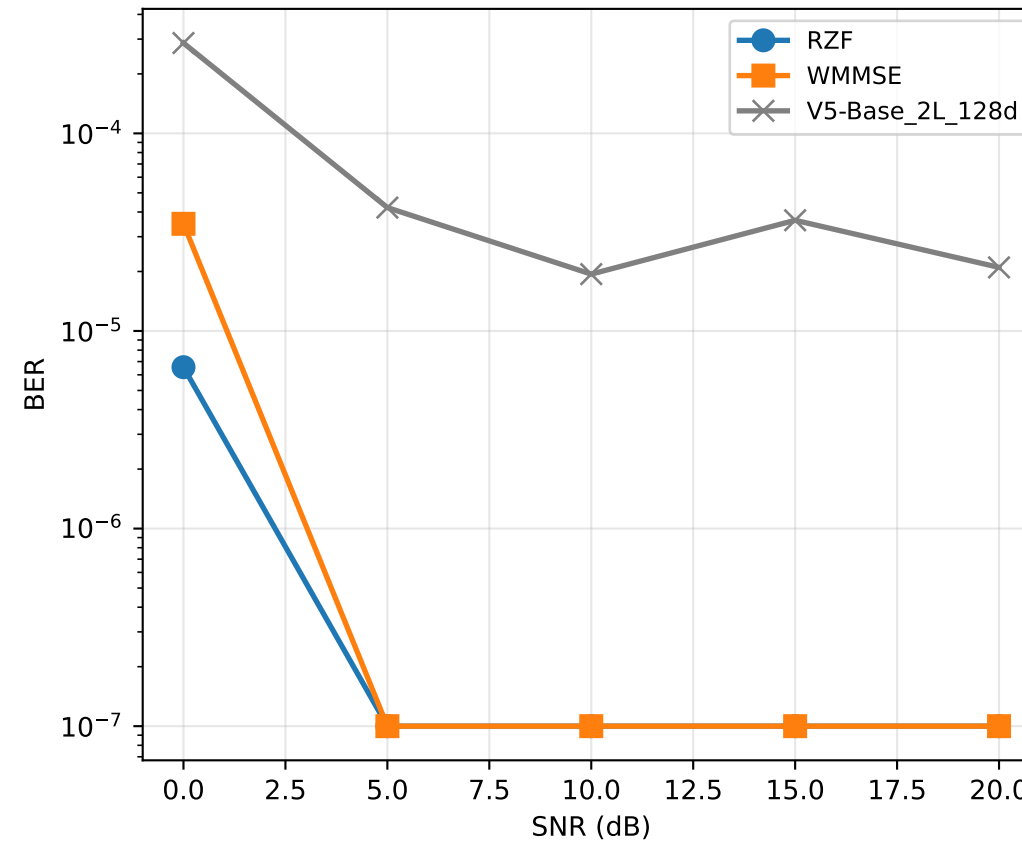


Transformer Precoding — V4 vs V5

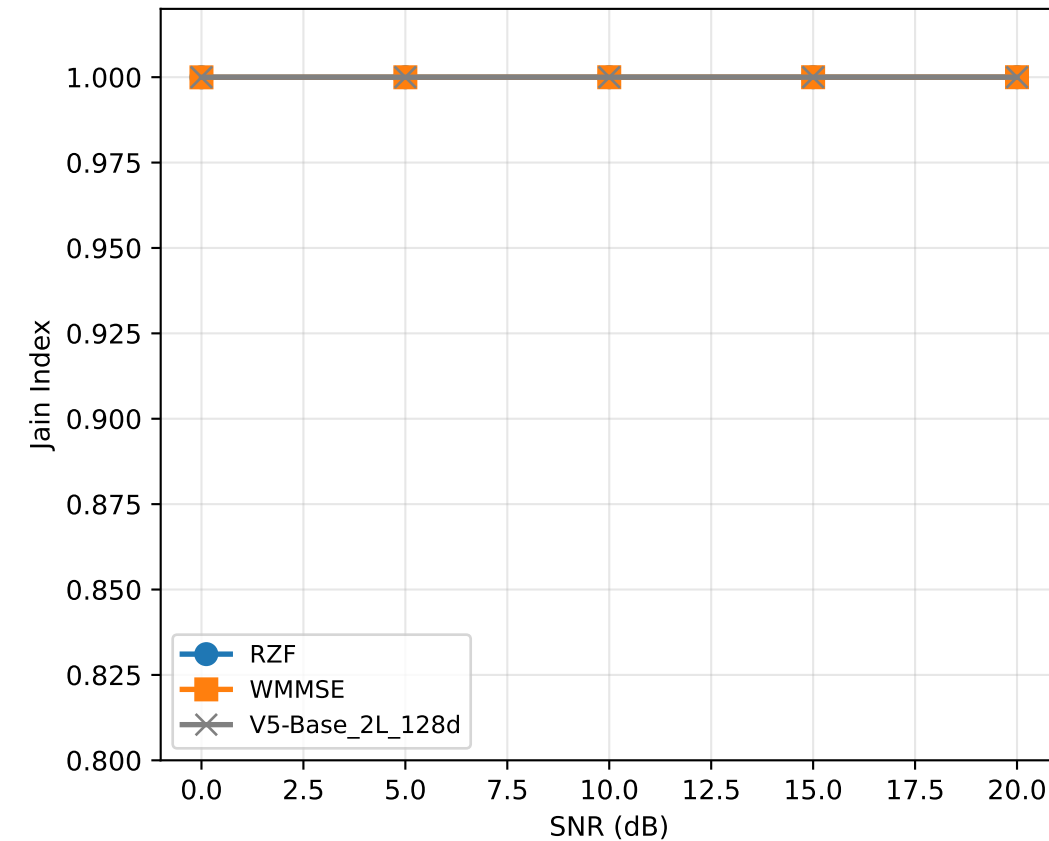
(a) Spectral Efficiency



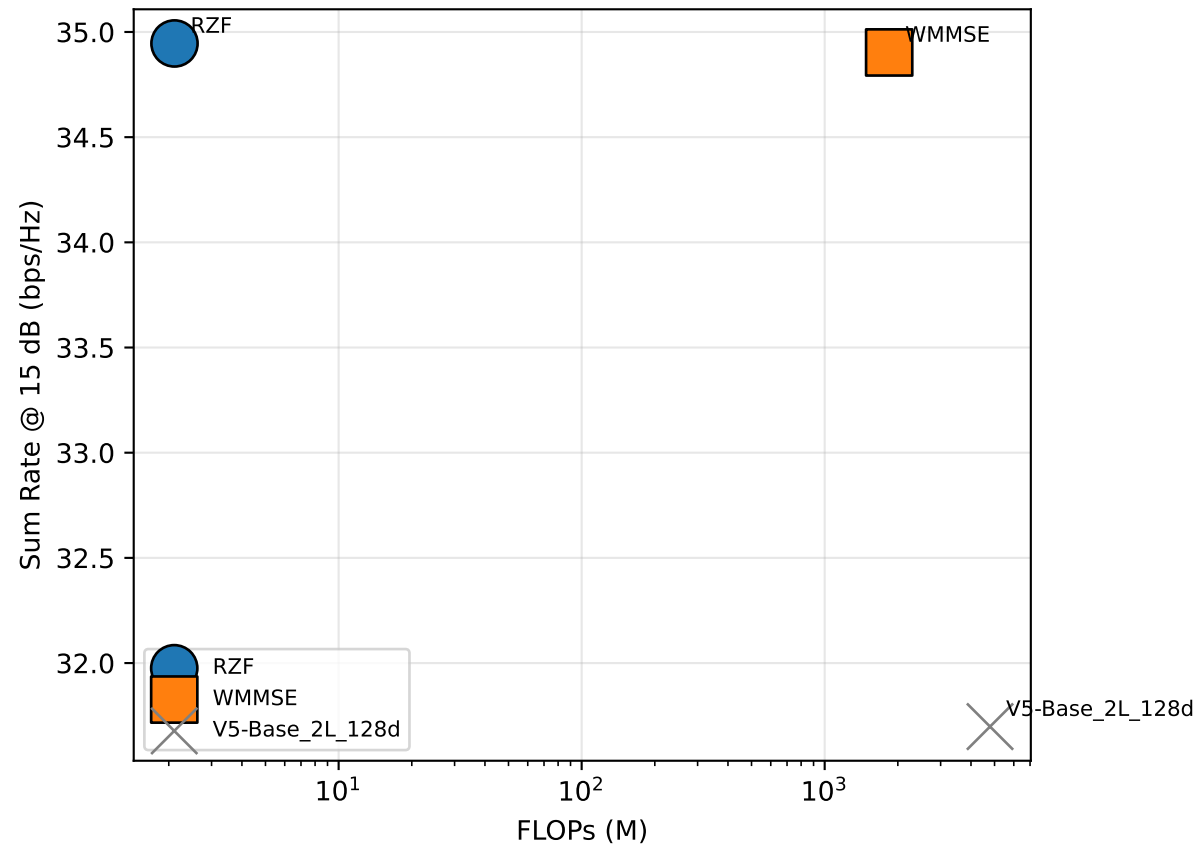
(b) Bit Error Rate



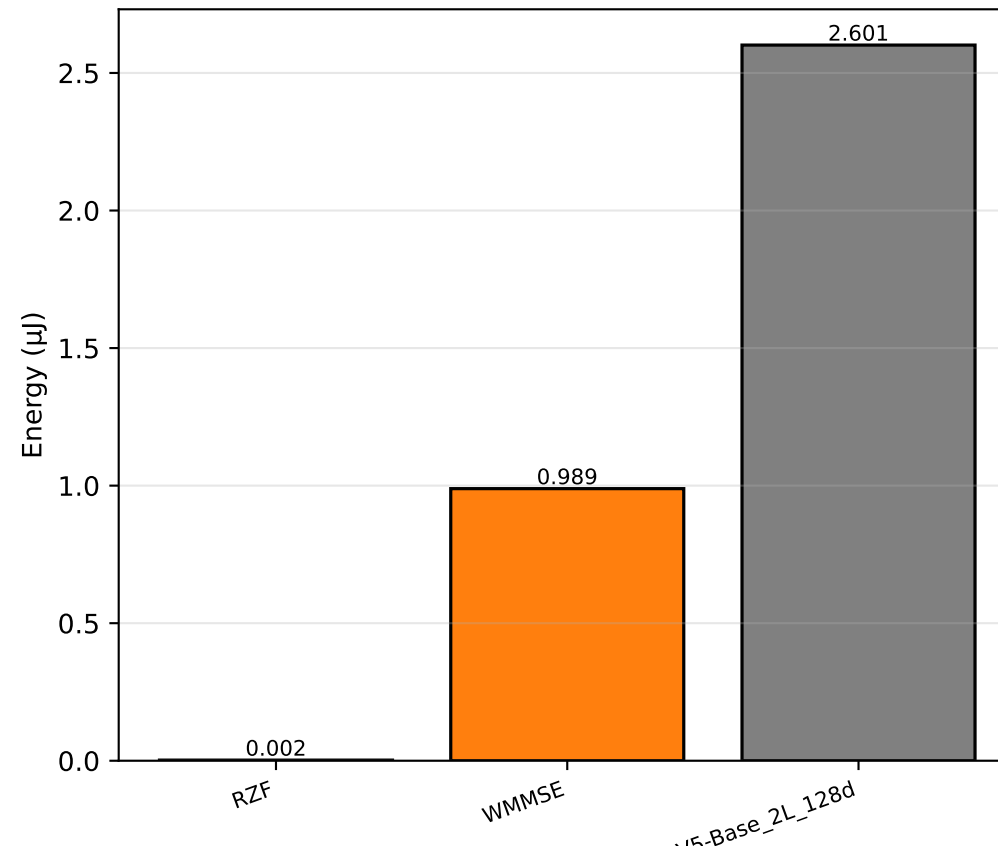
(c) User Fairness



(d) Complexity-Performance Pareto



(e) Inference Energy (μ J)



(f) Pareto @ 15 dB

Method	Rate	Gap	FLOPs	Enrgy
RZF	34.95	+0.04	2.1M	0.0023
WMMSE	34.90	+0.00	1842.5M	0.9891
V5-Base_2L_128d	31.70	-3.20	4791.3M	2.6014